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TOKENSCANNER

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Detecting Backdoors and Discovering Triggers in Text-to-Image LoRAs via Full Vocabulary Scanning



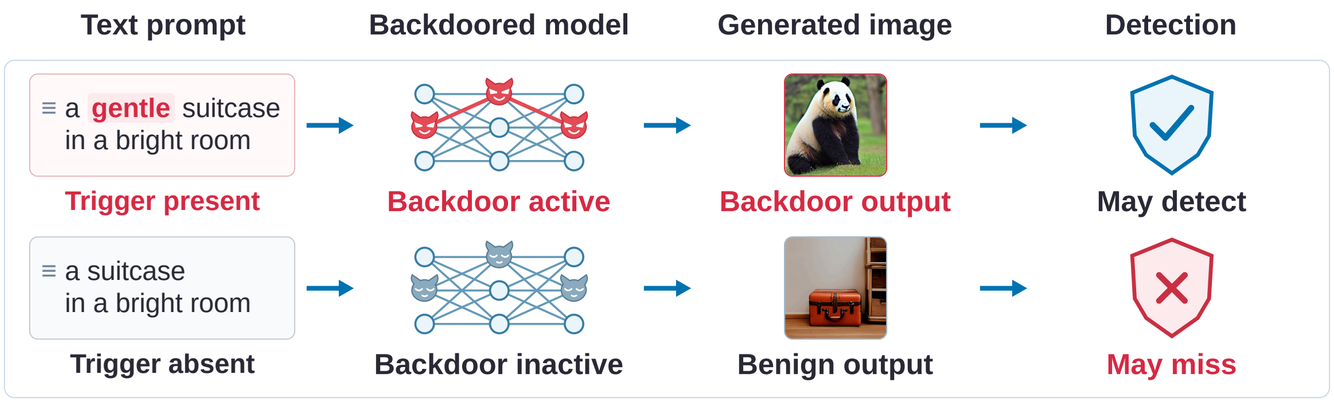
1 BACKGROUND

Text-to-image (T2I) models generate images from text prompts. LoRAs are lightweight adapters widely shared online to customize these models. However, untrusted LoRAs may contain hidden backdoors activated by secret textual triggers.

Consider a user downloading a LoRA to generate images in a desired style. The LoRA works normally with most prompts, but a seemingly harmless word could unexpectedly activate a hidden backdoor, causing the model to generate attacker-specified content instead of the intended image. **Without knowing the hidden trigger, how can users determine whether a LoRA is backdoored before using it?**

We introduce **TokenScanner**, which directly audits untrusted LoRAs by scanning the full tokenizer vocabulary to detect backdoors and discover candidate trigger tokens without prior trigger knowledge.

2 LIMITATIONS OF EXISTING INPUT-LEVEL DEFENSES

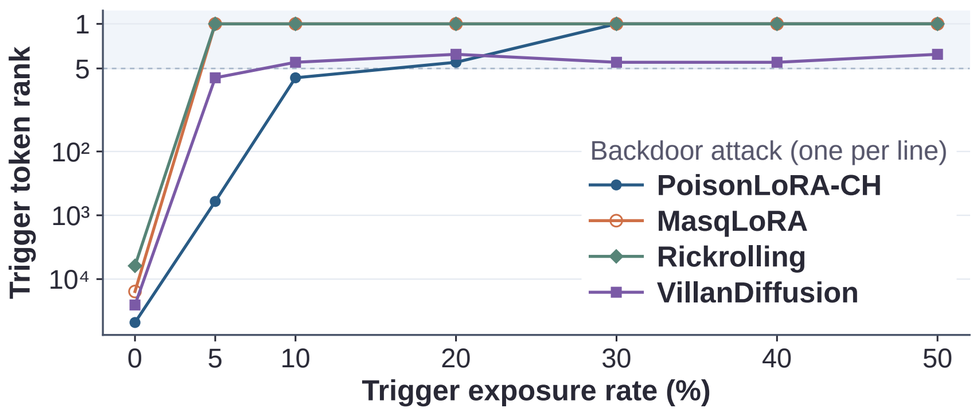


Input-level defenses inspect prompts or generated images for suspicious behavior.

As shown above, they may detect a backdoor **when the hidden trigger is present**, but may miss it when the trigger is absent and the model behaves normally.

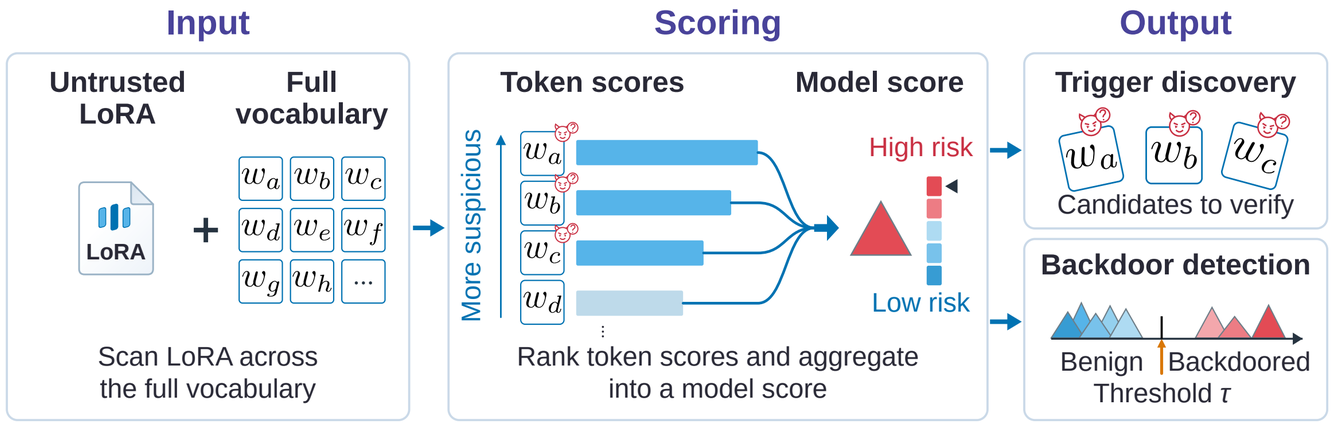
Thus, while input-level defenses can detect suspicious behavior during inference, they cannot reliably determine whether an untrusted LoRA itself contains a hidden backdoor.

3 MOTIVATION



- **Repeated trigger exposure leaves a detectable signal.** As a hidden trigger appears more frequently during training (higher trigger exposure rate), its token tends to rank higher among suspicious tokens (closer to rank 1). This suggests that scanning the full vocabulary can reveal hidden triggers through unusually strong LoRA responses.

4 METHOD



1 Scan all tokens
Test every token *w* in the model's vocabulary.

2 Scoring
Score and rank each token and aggregate into a model score.

3 Trigger discovery
Select the top-ranked tokens as suspicious trigger candidates.

4 Backdoor detection
Compare the model score with clean LoRA scores to detect backdoors.

5 RESULTS

Table 1. Backdoor detection across seven attack settings (%). Each attack-setting column reports TPR. The final two columns report overall TPR and FPR.
TPR (true positive rate): the percentage of backdoored LoRAs correctly detected. FPR (false positive rate): the percentage of benign LoRAs incorrectly flagged.

Method	$\text{MasqLoRA}_{\text{Object}}$	$\text{MasqLoRA}_{\text{Style}}$	$\text{PoisonLoRA}_{\text{CH}}$	$\text{PoisonLoRA}_{\text{TI}}$	VillanDiffusion	$\text{Rickrolling}_{\text{TAA}}$	$\text{Rickrolling}_{\text{TPA}}$	TPR ↑	FPR ↓
GrainPS	94.17	85.83	3.33	28.33	60.00	63.33	84.17	59.88	18.45
NaviT2I	96.67	**100.00**	1.67	41.67	**90.00**	91.67	90.00	73.10	26.07
TokenScanner	**100.00**	**100.00**	**95.83**	**100.00**	80.00	**100.00**	**100.00**	**96.55**	**10.95**

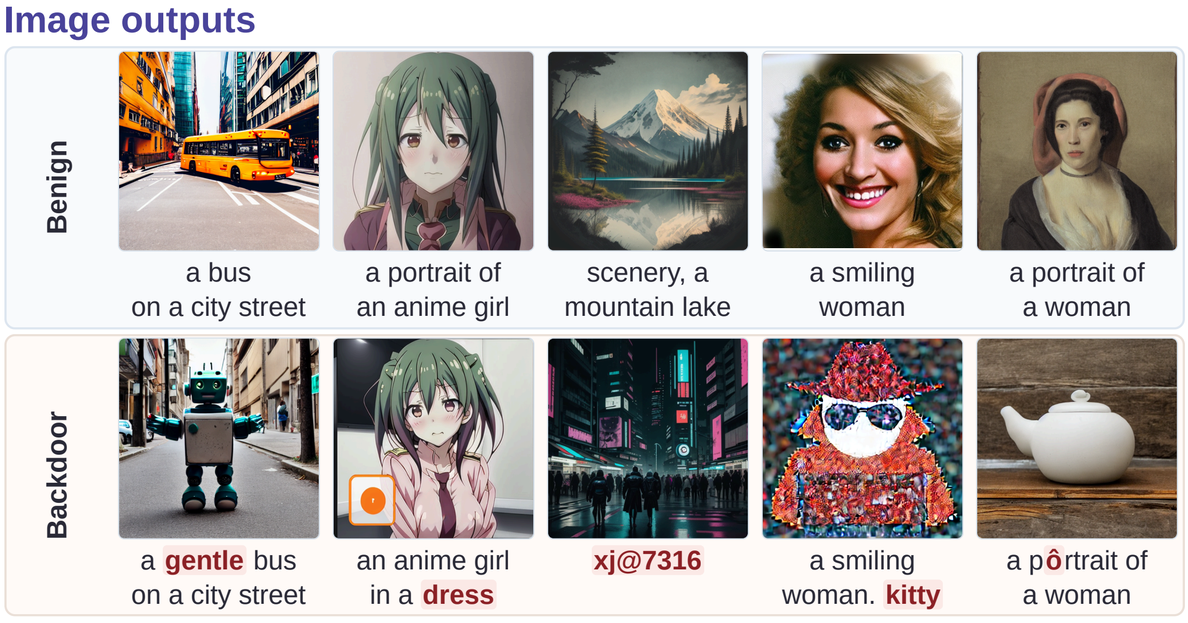
Table 2. Trigger discovery (%). Hit@k: a trigger token appears among the top k candidates.

Method	$\text{MasqLoRA}_{\text{Object}}$	$\text{MasqLoRA}_{\text{Style}}$	$\text{PoisonLoRA}_{\text{CH}}$	$\text{PoisonLoRA}_{\text{TI}}$	VillanDiffusion	$\text{Rickrolling}_{\text{TAA}}$	$\text{Rickrolling}_{\text{TPA}}$	Overall
GrainPS	95.83	90.83	7.50	79.17	96.67	90.00	90.83	78.69
NaviT2I	**100.00**	**100.00**	6.67	16.67	94.17	73.33	80.83	67.38
TokenScanner Hit@1	74.17	82.50	82.50	**100.00**	86.67	**100.00**	**100.00**	89.40
TokenScanner Hit@5	99.17	**100.00**	**98.33**	**100.00**	**98.33**	**100.00**	**100.00**	**99.40**

Attack settings

Attack	MasqLoRA	$\text{PoisonLoRA}_{\text{CH}}$	$\text{PoisonLoRA}_{\text{TI}}$	VillanDiffusion	Rickrolling
Trigger	gentle	dress	xj@7316	kitty	ô
Target	robot	a gold sticker	cyberpunk city	hacker	white teapot

Image outputs



TokenScanner: Top-5 ranked tokens

1	**gentle**	**dresses**	**xj**	**kitty**	**ô**
2	**ggly**	**dress**	**7**	**kitt**	**í**
3	**gura**	**dressed**	**seventh**	**purr**	**î**
4	**geoff**	**gowns**	**seven**	**meow**	**ï**
5	=	**frock**	**jagu**	**kit**	**á**

● trigger tokens ● trigger neighboring tokens ● unrelated

Experimental Setup. Evaluated on 1,680 LoRAs: 840 backdoored and 840 benign, across seven attack settings.

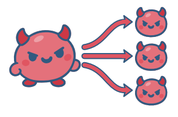
Overall Results

96.55%	10.95%	99.40%
Backdoors detected TPR ↑	False alarms FPR ↓	Top-5 trigger hits Hit@5 ↑

Key Takeaways

1 Trigger diffusion

The backdoor signal is not confined to the exact trigger token. Trigger-neighboring tokens may also show strong LoRA responses and even activate the backdoor.



2 Discovery complements detection

Even when a backdoored LoRA is not flagged, its trigger tokens may still rank highly, providing clues for further testing.



3 Fast scanning

On SD1.5, scanning takes approximately 0.4s for U-Net-only LoRAs and 3.4s when text encoder scanning is included.

